

Building this on Vespa — what transfers, and what to re-measure

This is a handover note for the team building library search on Vespa. It is not an argument about the engine: for a real service — hundreds of thousands of holdings, multiple branches, live circulation — Vespa is a reasonable choice and does several things this prototype had to hand-roll.

It is about everything the prototype **measured**, because that is the part a new engine does not give you and the part that is easy to lose in a rebuild. Some of it is counter-intuitive, and one finding runs directly against the thing Vespa makes easiest.

Every number here was produced by code in this repository and can be regenerated. Where a measurement is weak or contested, it says so.

1. What Vespa replaces, and what it does not decide

WHAT THE PROTOTYPE DOES	ON VESPA
BM25 in <code>src/bettersearch/keyword.py</code>	built in
Brute-force cosine over a numpy array	ANN with HNSW
Facet counting in <code>api/catalogue.py</code>	grouping
Fixed top-20 cut, then re-rank	phased ranking
<code>sentence-transformers</code> in the API process	an in-cluster embedder
A <code>pgvector</code> table as the production option	not needed

All of that is serving machinery, and Vespa does it better than we did. What Vespa does **not** decide — and what determined quality here — is the list below. Each has a measured answer, and each is a decision your schema and rank profiles will make implicitly whether or not anyone makes it deliberately:

- which encoder, and at what dimensionality
- what text actually gets embedded
- where the result list stops
- whether to fuse keyword and vector lanes, and how
- what to do about someone typing an exact title or author
- how you will know whether any of the above helped

2. The numbers, as an acceptance bar

41 queries over 4,000 real Open Library records, with 1,979 relevance judgements pooled from four retrieval lanes and judged blind. Each row names the encoder it used, because mixing them is how the gap gets misread:

ARM	NDCG@10
BM25 only	0.443
MiniLM 384d, semantic	0.579
RRF hybrid, BM25 + MiniLM	0.549 — below pure semantic
bge-base 768d, semantic	0.658 — what ships today
bge-base + LLM enrichment of thin records	0.743

A Vespa implementation over the same 4,000 records should reach **0.658** on semantic retrieval and **0.443** on BM25 alone. Below that, something in the port has gone wrong. Meaningfully above it, read §5 before celebrating.

Both figures were re-measured while writing this note and reproduce exactly — 0.6582 and 0.4433, at 100% judged coverage. The bar is real, not remembered.

The five deliberate known-item controls — exact titles and author names — score **1.000** today. They are a tripwire, not a target: if a change drops them, it has broken exact-name lookup, which is the one thing library users do most.

3. Four traps, each one hit or verified here

3.1 Hybrid is the easiest thing to switch on and the thing that lost

Reciprocal rank fusion of BM25 and vector scored **0.549 against pure semantic's 0.579** on this corpus. That result was initially explained away as an artefact of an eval set weighted towards keyword-hostile phrasing, with the prediction that real catalogue traffic — full of exact-name lookups — would be where fusion earned its place. It was re-tested on real records with five deliberate exact-name controls. It did not.

The mechanism is simple: blending two full rankings drags a good list down with a mostly-empty one, because BM25 returns nothing at all for a query that shares no term with any record, and a rank-based fusion still gives those empty positions weight.

This is **not** "do not use hybrid on Vespa". Vespa's phased ranking can express far better combinations than a flat RRF — a vector first phase with a BM25 signal among others in the second phase is a different and more promising shape. It is: **do not ship a hybrid rank profile because the engine offers one**. Measure it against a vector-only profile on the same judgements first.

3.2 Pooling strategy will silently halve your quality

Vespa's Hugging Face embedder defaults to `pooling-strategy: mean`. The `bge` family is trained for **CLS** pooling. Both our server and our browser build use CLS.

Leave the default with a bge model and nothing errors. The service starts, the queries return records, the facets work, and retrieval is quietly worse.

This is worth dwelling on because the same *shape* of failure already cost this project a week: the browser build ran a different encoder from the server, and the only symptom was "the standalone gives different results". There was no error, no warning, and no test that could have caught it, because both sides were

individually correct. Any mismatch between how documents and queries are encoded fails this way. Assume you will not notice it; add a check that would.

3.3 Circulation data must stay out of the embedded text

At real scale, availability, copies, due dates and branch change constantly. Bibliographic content does not.

Vespa supports partial updates to attribute fields that leave a tensor field on the same document untouched. Use that: **embed bibliographic content only**, and keep circulation state in attributes updated in place. If availability is part of the embedded text, every return and loan dirties a vector, and at a million holdings that is a re-embedding treadmill that will quietly become your largest compute cost.

The prototype arrives at the same place from a different direction: `content_hash` in `src/bettersearch/chunking.py` is taken over the embedded text only, so re-ingesting a corpus where only availability changed embeds nothing. That property is worth preserving explicitly in the schema rather than rediscovering.

3.4 Make the index carry its encoder's identity

Every index here stores the `model_id` it was built with and checks it on read *and* on write. Change the encoder and the index refuses to answer rather than scoring today's query against yesterday's vectors — which is the failure that returns confident, plausible, wrong results and raises nothing.

At a million records, re-embedding on a model change is a migration measured in hours, run alongside live traffic. The guard is what makes that survivable: whatever the Vespa equivalent is in your deployment — a field on the document, a rank profile that refuses an unexpected tensor dimension, a check in the feed pipeline — decide it before the first model change rather than during it.

4. What changes at real-library scale

The prototype is 4,000 records in one process. Things that did not matter there and will:

- **Cut depth becomes a phase boundary.** We cut the semantic list at a fixed top 20. That number was chosen by sweep — `scripts/tune_cutoff.py` — and fixed top-k beat every similarity threshold tried, because cosine has no absolute floor: on 4,000 records the threshold that looked right on 74 returned a median of 763 results. On Vespa this becomes `rerank-count` and the first phase/second phase split rather than a slice, but the finding stands: **do not cut a vector result list by score.**
- **Facets are counted before filtering.** Ours count over the whole ranked list so the sidebar shows what you *could* narrow to, not what survived the last click. Grouping makes this easy to get either way; it is a product decision worth making deliberately.
- **Multiple collections in one index.** We put 4,000 books and 114 events in one index and measured whether events were both findable and unobtrusive (`scripts/check_events.py`): 12/12 event-shaped queries reach an event, 0/5 known-item lookups show one. One index was enough. With more collection types — articles, archives, databases, e-resources — that balance needs re-checking, and rank fusion across collections is the lever if it tips.
- **Per-branch availability** is the obvious sharding and filtering question and the prototype says nothing useful about it. Ours is invented data.

5. How you will know you are better — and the trap in it

Read this before quoting any score, including ours.

The judgements were produced by pooling: the top 20 from each of four lanes — BM25, MiniLM over thin records, MiniLM over enriched records, and bge-base — unioned and judged blind. A record that no lane retrieved was never shown to a judge, and **an unjudged record counts as irrelevant**.

A Vespa implementation is a *fifth lane*. Every good record it finds that those four missed scores zero. **A better system can therefore score worse.**

This is not hypothetical and not small. It happened twice here:

- bge-base first measured *worse* than MiniLM, 0.578 against 0.588, and was nearly rejected. 100% of MiniLM's top 10 had been judged against **71%** of bge's. Re-pooling with bge as a fourth lane added 312 records no other lane had surfaced and moved it **+0.080** — the whole difference between "slightly worse" and "clearly better".
- A BM25 title-weighting change looked harmful for the same reason, and the sweep now reports coverage beside the score so nobody draws that conclusion again.

And it is reproducible in two minutes, which is the clearest way to see it. Adding one more collection to the index — 114 community events alongside the 4,000 books — produces this:

INDEX	NDCG@10, SEMANTIC	JUDGED COVERAGE
4,000 books (the pooled corpus)	0.6582	100%
the same books + 114 events	0.6511	97.6%

The score fell by 0.007. But coverage fell too, because event records were never in the pool and every one that surfaces is scored as irrelevant by definition. Some of that 0.007 is the events genuinely displacing books, and some is an artefact of the judgements — **and this eval cannot tell you which**. On the keyword lane the same move takes coverage from 100% to 89.3%.

That is a 114-record change. A new engine over a real catalogue will move coverage far more than that.

So: **scripts/score_rankings.py reports judged coverage next to every score.**

```
python scripts/score_rankings.py vespa-semantic.json
python scripts/score_rankings.py ours-semantic.json vespa-semantic.json # compared
```

It takes the simplest thing any engine can emit — a JSON object mapping each query to its ranked `doc_ids` — and reports `nDCG@10`, `Recall@5`, `MRR@10`, the known-item controls, events intrusion, the weakest queries, and coverage. It uses the same metric functions as everything else in this repository, so a Vespa number and ours are the same arithmetic by construction rather than by agreement.

Reading the coverage line:

COVERAGE OF YOUR TOP 10	WHAT IT MEANS
≥ 90%	the comparison is fair; believe the score
75–90%	treat with caution; the score is probably an undercount
< 75%	not comparable — you are measuring the pool, not the engine

If coverage comes back low, the honest move is to re-pool and re-judge rather than accept the number. `scripts/build_eval_set.py` does this and takes a `--lane` per system, so Vespa can be added as a lane and the whole set re-judged for a few pounds of model time.

What it cannot tell you: anything about latency, cost, freshness, or behaviour at a scale 4,000 records cannot reach. It is a relevance bar and only that.

6. Things worth not rebuilding

Two small pieces earned their place for reasons that are not about ranking, and are cheap to carry over:

- **Exact-match promotion.** A model-free pass over title and author that catches what neither lane can see — word order. It is worth almost nothing as ranking (nDCG 0.658 either way) and ships anyway, because it makes a *guarantee*: a record whose exact title someone typed is on page one because a rule says so, not because the encoder agreed. It also gives the card something true to say — "by this author" is legible in a way a cosine score is not.
 - **Saying why a record is there.** Keyword results name the query words the record does *not* contain; meaning-based results name the record's own subject headings closest to the query. Both are cheap, and they are most of what makes a meaning-based result list feel trustworthy rather than arbitrary.
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7. Open questions for your team

1. What is the actual corpus — holdings only, or articles, archives and e-resources too? The one-index finding was measured on two collections, not five.
2. Is enrichment of thin records on the table? It is worth +0.085 nDCG on top of the best encoder, and it is the second thing to try, not the first — a bigger encoder buys most of what it buys, for free.
3. What is the re-embedding plan when the encoder changes, and who notices if documents and queries fall out of step?
4. Will you re-pool the judgements with Vespa as a lane before comparing? If not, §5 says what the comparison is worth.
5. Multilingual? Every measurement here is English-only and the encoder choice would change.